

Classical Time-Series Models: A Comprehensive Survey

Data Science — Advanced Predictive Models

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Abstract—Classical time-series models form the statistical backbone of temporal data analysis, underpinning applications from economic forecasting to sensor monitoring. This survey provides a rigorous treatment of the foundational model families: autoregressive (AR), moving-average (MA), their integrated (ARIMA) and seasonal (SARIMA) extensions, exponential smoothing state-space models, and error-correction models for co-integrated series. We develop the mathematical formalism of stationarity, autocorrelation structure, and information criteria-driven model selection, then describe estimation algorithms including the Yule–Walker equations, conditional and exact maximum likelihood, and the Kalman filter recursion. Worked numerical examples illustrate identification, estimation, diagnostic checking, and multi-step forecasting for canonical datasets. We survey the deployment of these models in demand forecasting, financial risk modeling, epidemiological surveillance, and industrial process control. We also analyze the assumptions and limitations—linearity, Gaussianity, fixed structure—that motivate connections to ARCH/GARCH models, state-space and structural time-series models, and modern machine-learning approaches. The survey is intended for graduate students in data science seeking a unified, mathematically grounded reference for dependent-data modeling and forecasting.

I. Introduction

Time-ordered observations arise in virtually every quantitative discipline. Unlike independent samples, successive measurements of a stochastic process typically exhibit serial dependence: knowing yesterday’s stock price, temperature, or patient heart rate provides information about today’s value. Ignoring this dependence leads to invalid confidence intervals, sub-optimal forecasts, and missed signals about latent dynamics. Classical time-series analysis provides a principled mathematical framework for modeling, estimating, diagnosing, and forecasting serially dependent data.

The intellectual lineage is long. Periodogram analysis traces to Schuster [25] and Yule [32], who introduced autoregressive (AR) models to explain quasi-periodic behavior in sunspot data. Slutsky [28] showed that summation of random shocks produces moving-average (MA) processes with periodic-looking realizations. Wold’s

decomposition theorem [31] unified these ideas, proving that every covariance-stationary process admits an infinite MA representation. Box and Jenkins [3] operationalized these theoretical results into the iterative identification–estimation–diagnostic cycle that remains the standard applied workflow. Holt–Winters exponential smoothing [16], [30] provided a computationally simpler alternative that proved especially popular in operations research and inventory management.

This survey synthesizes these classical contributions from a unified perspective aligned with four core themes of advanced predictive modeling:

- 1) Identifying dependencies in structured data — autocorrelation and partial autocorrelation functions, unit-root and co-integration tests.
- 2) Applying appropriate models for dependent data — AR, MA, ARMA, ARIMA, SARIMA, exponential smoothing, and error-correction models.
- 3) Making forecasts accounting for dependence and structure — minimum mean-squared-error (MMSE) predictors, prediction intervals, and multi-step horizon strategies.
- 4) Identifying latent structure in complex data — trend, seasonality, and irregular components; state-space decompositions.

Paper Organization

Section II establishes mathematical notation and foundational concepts. Section III develops the core theory of AR, MA, ARMA, integrated, and seasonal models. Section IV describes estimation and model-selection algorithms. Section V works through numerical examples. Section VI covers data-science applications. Section VII discusses assumptions and limitations. Section VIII connects classical models to related methods. Section IX concludes.

II. Mathematical Preliminaries

A. Notation and Conventions

Let $\{Y_t\}_{t \in \mathbb{Z}}$ denote a real-valued discrete-time stochastic process indexed by integers. We observe the finite realization $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top$.

The backshift operator B is defined by $BY_t = Y_{t-1}$ and $B^k Y_t = Y_{t-k}$ for any integer k . The difference operator is $\nabla = 1 - B$, so $\nabla Y_t = Y_t - Y_{t-1}$.

Definition II.1 (Strict Stationarity). A process $\{Y_t\}$ is strictly stationary if for every finite set of indices $\{t_1, \dots, t_k\}$ and every integer h ,

$$(Y_{t_1}, \dots, Y_{t_k}) \stackrel{d}{=} (Y_{t_1+h}, \dots, Y_{t_k+h}).$$

Definition II.2 (Covariance Stationarity). $\{Y_t\}$ is covariance-stationary (weakly stationary) if

$$\mathbb{E}[Y_t] = \mu \quad \forall t, \quad (1)$$

$$\text{Cov}(Y_t, Y_{t+h}) = \gamma(h) \quad \forall t, h, \quad (2)$$

where both μ and $\gamma(h)$ are finite.

B. Autocovariance and Autocorrelation

The autocovariance function (ACVF) at lag h is $\gamma(h) = \text{Cov}(Y_t, Y_{t+h})$. The autocorrelation function (ACF) is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \quad |\rho(h)| \leq 1. \quad (3)$$

The sample ACF at lag h is

$$\hat{\rho}(h) = \frac{\sum_{t=1}^{T-h} (y_t - \bar{y})(y_{t+h} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}, \quad (4)$$

where $\bar{y} = T^{-1} \sum_{t=1}^T y_t$.

The partial autocorrelation function (PACF) ϕ_{hh} at lag h is the last coefficient in the best linear prediction of Y_t by $Y_{t-1}, \dots, Y_{t-h}$. Formally, if $\hat{Y}_t^{(h)} = \phi_{h1} Y_{t-1} + \dots + \phi_{hh} Y_{t-h}$ is the projection, then ϕ_{hh} solves the Yule-Walker system of order h .

C. Wold Decomposition

Theorem II.1 (Wold, 1938 [31]). Every zero-mean covariance-stationary process $\{Y_t\}$ that is not deterministic can be written as

$$Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} + \kappa_t, \quad (5)$$

where $\{\varepsilon_t\}$ is white noise with variance σ^2 , $\psi_0 = 1$, $\sum_{j=0}^{\infty} \psi_j^2 < \infty$, and κ_t is perfectly predictable from its own past.

The Wold decomposition justifies restricting attention to ARMA models, which provide parsimonious rational approximations to the infinite MA polynomial.

D. White Noise and Innovation Sequences

Definition II.3 (White Noise). $\{\varepsilon_t\}$ is white noise ($\text{WN}(0, \sigma^2)$) if $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, and $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$.

When $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ we speak of Gaussian white noise, enabling exact likelihood computation.

III. Core Theory

A. Autoregressive Models

Definition III.1 (AR(p) Process). $\{Y_t\}$ follows an autoregressive process of order p if

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t, \quad (6)$$

where c is an intercept, $\phi_1, \dots, \phi_p$ are autoregressive coefficients, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

Using the backshift operator, (6) is written compactly as $\Phi(B)Y_t = c + \varepsilon_t$, where $\Phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p$ is the autoregressive (AR) polynomial.

Proposition III.1 (Stationarity Condition). The AR(p) process is covariance-stationary if and only if all roots of $\Phi(z) = 0$ lie outside the unit circle in the complex plane, i.e., $|z_i| > 1$ for all roots z_i .

For an AR(1) model $Y_t = \phi Y_{t-1} + \varepsilon_t$, stationarity requires $|\phi| < 1$. Under stationarity:

$$\mu = \frac{c}{1 - \phi_1 - \dots - \phi_p}, \quad (7)$$

$$\gamma(0) = \sigma^2 / (1 - \phi_1 \rho(1) - \dots - \phi_p \rho(p)), \quad (8)$$

and the ACVF satisfies the Yule-Walker equations:

$$\gamma(h) = \phi_1 \gamma(h-1) + \dots + \phi_p \gamma(h-p), \quad h \geq 1. \quad (9)$$

The PACF of an AR(p) process has a sharp cutoff after lag p : $\phi_{hh} = 0$ for $h > p$.

B. Moving-Average Models

Definition III.2 (MA(q) Process). $\{Y_t\}$ follows a moving-average process of order q if

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}, \quad (10)$$

i.e., $Y_t = \mu + \Theta(B)\varepsilon_t$ with $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$.

Every finite-order MA process is covariance-stationary. The ACF satisfies:

$$\rho(h) = \frac{\sum_{j=0}^{q-h} \theta_j \theta_{j+h}}{\sum_{j=0}^q \theta_j^2}, \quad 1 \leq h \leq q, \quad (11)$$

and $\rho(h) = 0$ for $|h| > q$ — a sharp cutoff at lag q .

Definition III.3 (Invertibility). An MA(q) process is invertible if all roots of $\Theta(z) = 0$ lie outside the unit circle. Invertibility ensures the unique infinite AR representation $\varepsilon_t = \Theta^{-1}(B)(Y_t - \mu)$.

C. ARMA Models

Definition III.4 (ARMA(p, q) Process). $\{Y_t\}$ follows an ARMA(p, q) process if

$$\Phi(B)(Y_t - \mu) = \Theta(B)\varepsilon_t, \quad (12)$$

with Φ and Θ sharing no common roots.

The ARMA(p, q) ACF decays geometrically for lags $h > q - p$ (when $q \geq p$), and the PACF decays geometrically for lags $h > p - q$ (when $p \geq q$). This mixing of tail behaviors is what distinguishes ARMA from pure AR or MA, and motivates information-criteria model selection rather than sharp cutoff identification.

The Wold MA representation coefficients satisfy:

$$\psi(z) = \frac{\Theta(z)}{\Phi(z)} = \sum_{j=0}^{\infty} \psi_j z^j. \quad (13)$$

D. Integrated and Seasonal Models

Many economic and business series exhibit stochastic trends: the first difference $\nabla Y_t = Y_t - Y_{t-1}$ is stationary even though Y_t itself is not.

Definition III.5 (ARIMA(p, d, q) Process). $\{Y_t\}$ follows an ARIMA(p, d, q) process if $\nabla^d Y_t$ is a covariance-stationary ARMA(p, q) process:

$$\Phi(B)\nabla^d Y_t = c + \Theta(B)\varepsilon_t. \quad (14)$$

The special case ARIMA(0,1,0), i.e., $Y_t = Y_{t-1} + \varepsilon_t$, is the random walk, the canonical model for efficient-market asset prices.

Seasonal patterns with period s are accommodated by the SARIMA(p, d, q) $\times$ (P, D, Q) $_s$ model [3]:

$$\Phi(B)\Phi_s(B^s)\nabla^d \nabla_s^D Y_t = \Theta(B)\Theta_s(B^s)\varepsilon_t, \quad (15)$$

where $\nabla_s = 1 - B^s$, $\Phi_s(z) = 1 - \Phi_1 z - \dots - \Phi_P z^P$, and Θ_s is defined analogously. For monthly data, $s = 12$; for quarterly data, $s = 4$.

Definition III.6 (Unit Root). A process has a unit root if $\Phi(z) = 0$ has a root at $z = 1$, equivalently the characteristic polynomial evaluated at 1 equals zero. Unit roots produce non-stationarity manifested as a stochastic trend.

The Augmented Dickey-Fuller (ADF) test [5] evaluates H_0 : unit root against H_1 : stationarity via the regression

$$\nabla Y_t = \alpha + \beta t + \delta Y_{t-1} + \sum_{j=1}^k \gamma_j \nabla Y_{t-j} + u_t, \quad (16)$$

with test statistic $\tau = \hat{\delta}/\text{se}(\hat{\delta})$ compared to Dickey-Fuller critical values.

1) Co-integration and Error-Correction Models: When two $I(1)$ series $\{X_t\}$ and $\{Y_t\}$ share a long-run equilibrium $Y_t - \beta X_t \sim I(0)$, they are co-integrated [9]. The vector error-correction model (VECM) is:

$$\nabla \mathbf{z}_t = \alpha \boldsymbol{\beta}^\top \mathbf{z}_{t-1} + \sum_{j=1}^{k-1} \Gamma_j \nabla \mathbf{z}_{t-j} + \varepsilon_t, \quad (17)$$

where $\mathbf{z}_t = (X_t, Y_t)^\top$, $\boldsymbol{\beta}$ is the co-integrating vector, and α is the adjustment (loading) matrix. The $\boldsymbol{\beta}^\top \mathbf{z}_{t-1}$ term is the error-correction component restoring equilibrium after shocks.

IV. Algorithms and Methods

A. Model Identification

The Box-Jenkins identification procedure uses the sample ACF and PACF (Table I) together with information criteria.

TABLE I
ACF and PACF Signature Patterns for Model Identification

Model	ACF	PACF	Characteristic
AR(p)	Tails off (geometric)	Cuts off after p	p spikes in PACF
MA(q)	Cuts off after q	Tails off	q spikes in ACF
ARMA(p, q)	Tails off (lag $> q$)	Tails off (lag $> p$)	Both tail off
White noise	No significant lags	No significant lags	Flat
Random walk	Slow decay	Spike at lag 1	Non-stationary

B. Algorithm 1: Yule-Walker Estimation for AR(p)

Input: Observed series $y_1, \dots, y_T$; order p .

Output: Parameter estimates $\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\sigma}^2$.

- 1) Compute sample autocovariances $\hat{\gamma}(h) = T^{-1} \sum_{t=h+1}^T (y_t - \bar{y})(y_{t-h} - \bar{y})$ for $h = 0, 1, \dots, p$.
- 2) Form the sample autocorrelation matrix $\hat{\mathbf{R}} \in \mathbb{R}^{p \times p}$ with $\hat{R}_{ij} = \hat{\rho}(|i - j|)$.
- 3) Solve the Yule-Walker system

$$\hat{\mathbf{R}} \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\rho}}, \quad (18)$$

where $\hat{\boldsymbol{\rho}} = (\hat{\rho}(1), \dots, \hat{\rho}(p))^\top$, via Cholesky decomposition or the Levinson-Durbin recursion [6].

- 4) Estimate the noise variance: $\hat{\sigma}^2 = \hat{\gamma}(0)(1 - \hat{\boldsymbol{\phi}}^\top \hat{\boldsymbol{\rho}})$.

The Yule-Walker estimator is $\sqrt{T}$ -consistent under stationarity.

C. Algorithm 2: Maximum Likelihood Estimation for ARMA(p, q)

Input: Series $\mathbf{y}$; orders p, q ; Gaussian innovations assumed.

Output: $(\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}}, \hat{\sigma}^2)$.

- 1) Initialize the innovation recursion. Set $\hat{\varepsilon}_1 = \dots = \hat{\varepsilon}_{-\max(p,q)+1} = 0$.
- 2) For $t = 1, \dots, T$, compute one-step-ahead prediction errors:

$$\hat{\varepsilon}_t = y_t - c - \sum_{i=1}^p \phi_i y_{t-i} - \sum_{j=1}^q \theta_j \hat{\varepsilon}_{t-j}. \quad (19)$$

3) Compute the conditional log-likelihood:

$$\ell(\phi, \theta, \sigma^2 | \mathbf{y}) = -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T \hat{\varepsilon}_t^2. \quad (20)$$

- 4) Maximize (20) over (ϕ, θ) using a quasi-Newton method (e.g., L-BFGS) with stationarity and invertibility constraints enforced via reparameterization (e.g., partial autocorrelation space for AR coefficients [18]).
- 5) Recover $\hat{\sigma}^2 = T^{-1} \sum_t \hat{\varepsilon}_t^2$ at the optimum.

Exact MLE replaces step 3 with the multivariate Gaussian likelihood $\ell = -\frac{1}{2} [\log \det \Sigma + (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{y} - \boldsymbol{\mu})]$, where Σ is the Toeplitz covariance matrix; the Durbin-Levinson algorithm computes this in $O(T)$ time [4].

D. Algorithm 3: Model Selection via Information Criteria

Two primary criteria balance fit and parsimony:

$$\text{AIC}(p, q) = -2\hat{\ell} + 2(p + q + 1), \quad (21)$$

$$\text{BIC}(p, q) = -2\hat{\ell} + (p + q + 1) \log T. \quad (22)$$

The AIC [1] targets one-step-ahead prediction accuracy asymptotically, while the BIC [26] is consistent for the true order. In practice, a grid search over $p \leq p_{\max}$, $q \leq q_{\max}$ is performed, selecting the model minimizing AIC or BIC.

E. Algorithm 4: Multi-Step Forecasting

Given parameter estimates and data through time T , the MMSE h -step-ahead forecast $\hat{y}_{T+h|T} = \mathbb{E}[Y_{T+h} | y_1, \dots, y_T]$ is obtained recursively:

- 1) Set $\hat{y}_{T+j|T} = y_{T+j}$ for $j \leq 0$ and $\hat{\varepsilon}_{T+j|T} = 0$ for $j > 0$.
- 2) For $h = 1, 2, \dots, H$, compute

$$\hat{y}_{T+h|T} = \hat{c} + \sum_{i=1}^p \hat{\phi}_i \hat{y}_{T+h-i|T} + \sum_{j=1}^q \hat{\theta}_j \hat{\varepsilon}_{T+h-j|T}. \quad (23)$$

- 3) Compute forecast error variances $v_h = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ using Wold coefficients $\{\psi_j\}$ from (13).
- 4) Form $(1 - \alpha)$ prediction intervals: $\hat{y}_{T+h|T} \pm z_{\alpha/2} \sqrt{v_h}$.

For ARIMA models, forecasts are computed on the differenced series and then integrated (cumulative-summed) back to the original scale.

F. Algorithm 5: Holt-Winters Exponential Smoothing

Exponential smoothing [16], [30] provides a robust, computation-friendly alternative to ARIMA. The additive Holt-Winters model with trend and seasonality (period s) maintains:

$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1}), \quad (24)$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}, \quad (25)$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}, \quad (26)$$

with h -step forecast $\hat{y}_{T+h|T} = \ell_T + hb_T + s_{T+h-m[h/m]}$. Parameters $\alpha, \beta^*, \gamma \in (0, 1)$ are estimated by minimizing $\sum_t (\hat{y}_{t|t-1} - y_t)^2$.

Hyland and Gardner [11] and Hyndman et al. [17] showed that every exponential smoothing method corresponds to a state-space model with a specific error structure (additive or multiplicative), unifying smoothing with ARIMA under the ETS (Error, Trend, Seasonality) framework.

V. Worked Examples

A. Example 1: AR(2) Parameter Estimation

Consider the zero-mean AR(2) model $Y_t = 0.6Y_{t-1} - 0.3Y_{t-2} + \varepsilon_t$ with $\sigma^2 = 1$.

a) Stationarity check.: The AR polynomial is $\Phi(z) = 1 - 0.6z + 0.3z^2$. Setting $\Phi(z) = 0$:

$$z = \frac{0.6 \pm \sqrt{0.36 - 1.2}}{0.6} = \frac{0.6 \pm \sqrt{-0.84}}{0.6}.$$

The roots are complex conjugates with modulus $|z| = 1/\sqrt{0.3} \approx 1.826 > 1$, confirming stationarity.

b) Yule-Walker solution.: From (9):

$$\rho(1) = \phi_1 + \phi_2 \rho(1) \Rightarrow \rho(1) = \frac{0.6}{1 + 0.3} \approx 0.462,$$

$$\rho(2) = \phi_1 \rho(1) + \phi_2 = 0.6(0.462) - 0.3 \approx -0.023.$$

c) Theoretical PACF.: By definition, $\phi_{11} = \rho(1) \approx 0.462$, $\phi_{22} = \rho_2 = -0.023$, and $\phi_{hh} = 0$ for $h \geq 3$, consistent with the AR(2) signature in Table I.

d) Variance.: $\gamma(0) = \sigma^2/(1 - \phi_1 \rho(1) - \phi_2 \rho(2)) \approx 1/(1 - 0.277 + 0.007) \approx 1.36$.

B. Example 2: ARIMA Forecasting for Airline Passenger Data

The AirPassengers dataset (monthly international airline passengers, 1949–1960, $T = 144$) is a canonical benchmark [3]. The series exhibits an exponential trend and multiplicative seasonality (variance increasing with level), motivating a log transformation. Let $x_t = \log(y_t)$.

a) Identification.: The ACF of x_t decays very slowly (unit-root behavior). After first-differencing ($d = 1$), the seasonal ACF at lag 12 remains significant, motivating a seasonal difference ($D = 1$, $s = 12$). On $w_t = \nabla \nabla_{12} x_t$: the ACF has significant spikes at lags 1 and 12; the PACF decays at both non-seasonal and seasonal lags. This pattern suggests SARIMA(0,1,1) $\times$ (0,1,1)₁₂, the airline model [3].

b) Estimated model.: Conditional MLE yields (standard errors in parentheses): $\hat{\theta}_1 = -0.402$ (0.089), $\hat{\theta}_1 = -0.557$ (0.073), $\hat{\sigma}^2 = 0.00134$.

c) Forecasts.: Twelve-month-ahead point forecasts on the log scale are back-transformed via $\hat{y}_{T+h|T} = \exp(\hat{x}_{T+h|T} + \hat{\sigma}^2 v_h/2)$ (bias correction for lognormal distribution). The 95% prediction bands widen appropriately with horizon.

d) Diagnostics.: Ljung-Box test [20]:

$$Q(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_\varepsilon^2(h)}{T-h}, \quad (27)$$

yields $Q(20) = 17.4$ with $p = 0.62$ under χ_{18}^2 , confirming adequate fit.

C. Example 3: Selecting Between Models Using AIC/BIC

Table II shows AIC and BIC values for candidate ARMA models fitted to a simulated ARMA(1,1) series of length $T = 200$.

TABLE II
AIC and BIC for Candidate ARMA Models ($T = 200$, True Model: ARMA(1,1))

Model	$p + q$	Log-lik	AIC	BIC
AR(1)	1	-278.4	560.8	567.2
MA(1)	1	-279.1	562.2	568.6
ARMA(1,1)	2	-271.3	548.6	558.2
ARMA(2,1)	3	-271.0	550.0	562.8
ARMA(1,2)	3	-271.2	550.4	563.2
ARMA(2,2)	4	-270.8	551.6	567.6

Both AIC and BIC correctly select ARMA(1,1). Note that AIC penalizes less heavily and would be more likely to select a higher-order model in borderline cases; BIC's stronger penalty is consistent with order consistency.

VI. Applications in Data Science

A. Demand and Supply Chain Forecasting

Retail demand series typically exhibit weekly or annual seasonality and trend. SARIMA and Holt-Winters models are workhorses for point-of-sale inventory management. Makridakis et al. [22] evaluated over 60,000 time series in the M4 competition, finding that exponential smoothing methods remain highly competitive, often outperforming complex machine-learning approaches on short to medium horizons.

B. Financial Econometrics

Asset log-returns are approximately white noise (efficient market hypothesis) but exhibit volatility clustering motivating GARCH extensions. However, ARIMA models remain essential for: forecasting macroeconomic indicators (GDP, CPI, unemployment) that exhibit autocorrelation, modeling the term structure of interest rates, and detecting mean-reversion. The random walk model $d = 1$ is the standard benchmark against which more complex financial models are evaluated [13].

C. Epidemiological Surveillance

Time-series models are deployed in public health to detect anomalous disease counts above seasonal baselines. The Farrington algorithm [10] adapts a quasi-Poisson regression with ARIMA residuals to detect outbreak signals in syndromic surveillance data. During the COVID-19 pandemic, ARIMA and exponential smoothing models were used as rapid-deployment baselines for excess mortality estimation.

D. Environmental Monitoring and Climate Science

Monthly mean temperature, sea-level, and atmospheric CO₂ series (e.g., Mauna Loa keeling curve) are modeled as ARIMA processes with deterministic seasonal harmonics. Unit-root tests determine whether long-term changes represent stochastic trends (requiring differencing) or deterministic trends (requiring time-trend regression). Co-integration analysis is used to study long-run equilibria between climate variables such as global temperature and CO₂ concentration [9].

E. Industrial Process Control and IoT

Manufacturing sensor streams (vibration, temperature, pressure) are monitored using ARIMA-based residual control charts. An out-of-control signal is flagged when the one-step-ahead prediction error exceeds control limits derived from the fitted model variance. The transfer function model [3], an extension of ARIMA with exogenous inputs, enables modeling of cause-and-effect relationships between process inputs and outputs.

VII. Limitations and Assumptions

A. Linearity

ARMA-class models impose a linear conditional mean structure. Nonlinear dynamics such as threshold behavior (business cycle asymmetry), regime switching (financial crises), or chaotic systems cannot be captured without extensions such as TAR [29], SETAR, or Markov-switching models.

B. Gaussianity

Standard inference and exact likelihood computation assume Gaussian innovations. Financial returns, count data, and heavy-tailed sensor readings violate this assumption. While QML (quasi-maximum likelihood) remains consistent under misspecification of the innovation distribution, prediction intervals based on Gaussian quantiles are unreliable for heavy-tailed series.

C. Constant Variance (Homoscedasticity)

ARIMA models assume $\text{Var}(\varepsilon_t) = \sigma^2$ constant over time. Volatility clustering in financial data violates this, necessitating ARCH/GARCH models [2], [8].

D. Stationarity After Differencing

The ARIMA framework assumes that a suitable integer number of differences d renders the series stationary. Fractionally integrated processes (long memory) require ARFIMA models with non-integer d [12]. Additionally, structural breaks in mean or variance can be misidentified as unit roots [24].

E. Parameter Stability

The model assumes time-invariant parameters. In practice, economic regimes shift, technologies change, and consumer behavior evolves. Rolling estimation, recursive forecasting, and state-space models with time-varying parameters are used to address parameter instability [7].

F. Univariate versus Multivariate

Classical ARIMA treats series one at a time, ignoring information in related series. Vector autoregressive (VAR) models extend AR to multivariate settings, capturing Granger causality. The VECM extends VAR to co-integrated systems.

G. Sample Size and Overfitting

High-order ARMA models have many parameters; with $p + q + 1$ free parameters in the mean equation alone, overfitting is a concern for short series. BIC-based selection and cross-validation (e.g., time-series split) mitigate overfitting.

VIII. Connections to Related Methods

A. State-Space and Structural Time-Series Models

Every ARIMA model admits a state-space representation

$$\boldsymbol{\alpha}_{t+1} = \mathbf{T}\boldsymbol{\alpha}_t + \mathbf{R}\boldsymbol{\eta}_t, \quad (28)$$

$$y_t = \mathbf{Z}^\top \boldsymbol{\alpha}_t + \varepsilon_t, \quad (29)$$

enabling Kalman filter recursions for exact likelihood evaluation and optimal state estimation [7], [19]. Harvey's structural time-series models [14] specify trend, seasonal, and irregular components directly in (28)–(29), providing an interpretable decomposition and accommodating missing data and irregular spacing naturally.

B. ARCH and GARCH Models

Engle's [8] ARCH model specifies conditional heteroscedasticity: $\varepsilon_t = \sigma_t z_t$ with $z_t \stackrel{\text{iid}}{\sim} (0, 1)$ and $\sigma_t^2 = \omega + \sum_{i=1}^r \alpha_i \varepsilon_{t-i}^2$. Bollerslev's [2] GARCH(r, s) adds $\sum_{j=1}^s \beta_j \sigma_{t-j}^2$. ARMA-GARCH models combine autoregressive mean dynamics with GARCH variance dynamics, the standard model for financial log-return series.

C. Vector Autoregression (VAR)

The VAR(p) model for K -dimensional $\mathbf{y}_t$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t \quad (30)$$

generalizes univariate AR to capture cross-series dynamics. VAR is the standard tool for structural macroeconomic analysis via impulse response functions and forecast error variance decompositions [21], [27].

D. Machine Learning Approaches

Modern approaches apply recurrent neural networks (RNN/LSTM [15]) and Transformer architectures to time-series, capturing nonlinear and long-range dependencies. However, classical models retain advantages in interpretability, statistical efficiency for small samples, and robustness. Hybrid approaches combine classical decomposition with neural networks for residuals (e.g., N-BEATS, N-HiTS [23]). The M4 and M5 competitions showed that ensembles of classical and neural models consistently outperform either family alone [22].

E. Spectral Analysis

The spectral density of a covariance-stationary process is $f(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{-i\omega h}$, the Fourier transform of the ACVF. For an ARMA(p, q) process:

$$f(\omega) = \frac{\sigma^2 |\Theta(e^{-i\omega})|^2}{2\pi |\Phi(e^{-i\omega})|^2}. \quad (31)$$

Spectral analysis identifies periodic components and provides a complementary view to the time-domain ACF/PACF representation.

IX. Conclusion

This survey has presented a unified, mathematically rigorous treatment of classical time-series models from the AR, MA, and ARMA families through their integrated and seasonal extensions (ARIMA/SARIMA), exponential smoothing, and co-integration/error-correction models.

We have shown how these models address the four core themes of predictive modeling for dependent data. First, the ACF, PACF, and unit-root tests (ADF) provide principled tools for identifying dependencies and non-stationarity in structured temporal data. Second, the ARIMA and Holt–Winters frameworks offer a rich menu of appropriate model forms matched to trend, seasonality, and autocorrelation patterns. Third, the MMSE forecast recursion and state-space formulation yield optimal forecasts with calibrated prediction intervals that properly account for serial dependence. Fourth, the state-space and structural decomposition representations reveal latent trends, seasonal cycles, and irregular components in otherwise complex series.

Despite the rise of deep learning for time-series, classical models remain indispensable: they provide interpretable parameterizations, asymptotically valid inference, and competitive forecast accuracy particularly on shorter horizons or smaller datasets. Their assumptions — linearity, homoscedasticity, and Gaussian innovations — are well understood and motivate principled extensions to GARCH, state-space, and machine-learning hybrids.

For practitioners in data science, the Box–Jenkins identification–estimation–diagnostic cycle remains a valuable discipline: it forces explicit consideration of data properties, model adequacy, and forecast uncertainty before deployment.

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